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On the validation of the TDAC method for a stochastic turbulent combustion model in OpenFOAM

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Abstract

In simulations of reacting flows, it is common that the majority of computational effort is related to the integration of the ordinary differential equations (ODE) for chemistry. This present study aims to increase the efficiency of the chemical kinetic solution in an existing stochastic turbulent combustion solver through use of the Tabulation of Dynamic Adaptive Chemistry (TDAC) method which is a combination of the well-known in-situ adaptive tabulation (ISAT) approach and the dynamic adaptive chemistry (DAC) dimension reduction scheme. The accuracy and numerical parameter sensitivity of ISAT and DAC are tested individually for an experimental jet flame configuration, and then the best combination of parameter settings is selected for simulation using the combined TDAC method. Results show convergence with reducing tolerances and significant speedup is achieved.

1 Introduction

Combustion is a complex but also very common process in engineering applications and computational modelling is increasingly being used as a tool for combustor design. A key difficulty of such numerical work is the wide span of time scales and associated stiffness involved in describing the non-linear chemical kinetics and this poses prohibitive computational costs, especially for practical fuels with large molecular weights and their multiple, parallel detailed reaction pathways. Most practical combustion is turbulent, and this further increases computational cost due to the non-equilibrium chemical states that are produced by turbulence-chemistry interactions and due to the small length scales which are partially resolved in large eddy simulations (LES). Probability density function (PDF) models [1] (called filtered density function, FDF, in LES) are among the most sophisticated turbulent combustion approaches and have an enormous advantage over other models in that the non-linear chemistry source terms appear naturally in the transport equations in closed form without simplification. The PDF transport equation is highly dimensional and, in order to yield computationally tractable solutions, the partial differential equations are recast into equivalent stochastic differential equations (SDE) and solved on an ensemble of notional Lagrangian particles. However, within the PDF equation, a closure model is required for the dissipation of conditional variance and various mixing models have been proposed over the years. These models have desirable properties and key among these is compositional localness. The MMC model [2] enforces localness through an independent reference space which provides a quality closure and has led to the development of a highly computationally efficient sparse stochastic particle method called MMC-LES [3]. The model and an associated object orientated C++ opensource solver called *mmcFoam* [4] have been validated against a range of laboratory flames, including plane, inhomogeneous inlet, lifted, swirling and high-pressure jet flames [5-9], and flames involving nanoparticle synthesis [10].

Integration of the SDE's includes a fractional step to evolve the chemical state of each notional particle through solution of coupled ODE's for the kinetics of each species. The default approach is direct integration (DI) via stiff ODE solvers such as the Runge-Kutta or Rosenbrock methods [11]. In using *mmcFoam,* we have observed that the computational effort associated with DI can account for around half of the total computational load when using a relatively small kinetics scheme with 19 species [8]. A higher proportion may be expected for more detailed chemical schemes. As researchers turn towards simulation of practical fuels requiring many tens or hundreds of species, even efficient methods like sparse MMC-LES will become intractable and additional efficiency measures will be required. The objective of this study is to expedite the calculation of the chemistry in *mmcFoam* by incorporating the Tabulation of Dynamic Adaptive Chemistry (TDAC) method [12] which is a combination of in-situ adaptive tabulation (ISAT) [13] and the dynamic adaptive chemistry (DAC) reduction scheme [14]. TDAC exists within the latest OpenFOAM release (versions 5 and 6) but to the best of our knowledge this is its first application within a stochastic OpenFOAM-based solver. The performance is investigated in simulations of a piloted flame with a compositionally homogeneous inlet that was experimentally investigated at the University of Sydney [15]. Results for temperature, species and computational speedup are compared for simulations using ISAT and DAC alone and the combined TDAC method for simulations using three different chemical mechanisms ranging from one with 22 species through to one with 111 species.

2 Methodology

2.1 In-situ adaptive tabulation (ISAT)

ISAT stores the chemical states and reaction mapping computed by DI and, when possible, extrapolates them to approximate later variations in the chemical state. The chemical states are stored as leaves in a binary tree structure so that during subsequent retrievals the most similar leaf to the present state can be identified without scanning through all of them. New solutions by DI can be added to the tree, while existing leaves can either grow to encompass a larger region in composition space for valid retrieval or be deleted to free up memory if not being used for regular retrievals. The essential formulation of ISAT is provided below and complete details can be found in [13, 16-18].

For an initial thermochemical state $\varphi(t_0) = \varphi^0$, where $\varphi =$

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$\{\boldsymbol{Y}, T, P\}$ and $\boldsymbol{Y} = \{Y_1, Y_2, \ldots, Y_{n_s}\}$ is species mass fraction, T is temperature and P is pressure, the trajectory of $\varphi(t)$ in thermochemical state-space is determinant and unique. Thus, the reacted thermochemical state for a fixed timestep is a unique function of φ^0 expressed as $\varphi(t_0 + \Delta t) = \boldsymbol{R}(\varphi^0)$, which is called the reaction mapping. If a query point φ^q (i.e. a composition point whose reaction mapping is required) is similar to the φ^0 that are stored on leaves, DI for $\varphi^q(t_0 + \Delta t)$ is replaced by the linear approximation

$$\boldsymbol{R}(\varphi^q) \approx \boldsymbol{R}^l(\varphi^q) = \boldsymbol{R}(\varphi^0) + \delta\boldsymbol{R}^l \tag{8}$$

where $\delta\boldsymbol{R}^l = \boldsymbol{A}(\varphi^q - \varphi^0) = \boldsymbol{A}\delta\varphi$ and $\boldsymbol{A}$ is the stored mapping gradient matrix with $A_{ij} = \partial R_i(\varphi^0)/\partial\varphi_j$. We can define an error associated with the linear approximation given by Eq. (8) as

$$\varepsilon_{local} = \left|\boldsymbol{R}(\varphi^q) - \boldsymbol{R}^l(\varphi^q)\right|, \tag{9}$$

and compositions satisfying $\varepsilon_{local} \leq \varepsilon_{ISAT}$ are said to comprise a region of accuracy (ROA) within which the approximation is valid. For small $\delta\boldsymbol{R}^l$ the ROA approaches a hyper-ellipsoid [13], called the ellipsoid of accuracy (EOA), that is given by the regions satisfying the inequality

$$\delta\varphi^T \tilde{\boldsymbol{A}}^T \boldsymbol{B}^T \boldsymbol{B} \tilde{\boldsymbol{A}}\, \delta\varphi \leq \varepsilon_{ISAT}^2 \tag{10}$$

where $\tilde{\boldsymbol{A}}$ is the modification of $\boldsymbol{A}$ in order to avoid singular values and $\boldsymbol{B}$ is the scaling matrix as explained in [13]. At each timestep the following operations take place.

Retrieval: If Eq. (10) is satisfied for a query point φ^q, then the data is retrieved and the new composition, $\boldsymbol{R}(\varphi^q)$, is obtained by (8). Move to the next query point.

Direction integration: If Eq. (10) is not satisfied, then $\boldsymbol{R}(\varphi^q)$ is obtained by DI and ε_{local} is calculated according to Eq. (9). Move to either the growth or add steps.

Growth: If $\varepsilon_{local} \leq \varepsilon_{ISAT}$, the current EOA is deemed as being too conservative and the leaf is grown according to the minimum volume rule [17] to include the current query point. Move to next query point.

Add: If $\varepsilon_{local} > \varepsilon_{ISAT}$, a new leaf is added to the tree for possible use in later retrievals. Move to next query point.

Removal: To reduce memory usage leaves are removed from the tree if they have not been accessed in the last N_d time steps, where N_d has a user specified value.

2.2 Dynamic adaptive chemistry (DAC)

The DAC method [14] aims to locally (i.e. on each notional particle) and dynamically (i.e. temporally) reduce the number of chemically active species, thus reducing the size of coupled systems of ODE's and the computational cost of integrating them. It uses a reduction algorithm based on the direct relation graph with error propagation (DRGEP) concept [19]. Fig. 1a shows a DRG consisting of five species with some of the reactions, indicated by the arrows (called edges), being reversible. The strength of the kinetic relationships between the species is indicated schematically by the width of the edges and is defined by a contribution matrix

$$r_{AB} = \frac{\left|\sum_{i=1}^{n_r} \nu_{i,A}\omega_i\delta_B^i\right|}{\max\left(P_A, C_A\right)} \tag{11}$$

where

$$P_A = \sum_{i=1}^{n_r} \max\left(0, \nu_{i,A}\omega_i\right) \tag{12}$$

$$C_A = \sum_{i=1}^{n_r} \max\left(0, -\nu_{i,A}\omega_i\right) \tag{13}$$

In the above n_r is the number of reactions, $\nu_{i,A}$ is the stoichiometric coefficient of species A in the *i*th reaction, ω_i is the reaction rate, and δ_B^i equals one if the *i*th reaction involves species B and is otherwise equal to zero. To reduce the dimension of the composition space edges for which $r_{AB} < \varepsilon_{DAC}$ are eliminated, and the DRG is reconfigured as shown in Fig. 1b. Here ε_{DAC} is a specified DAC error tolerance. Species which are eliminated (i.e. species E in the figure) are chemically frozen but continue to participate in three-body reactions.

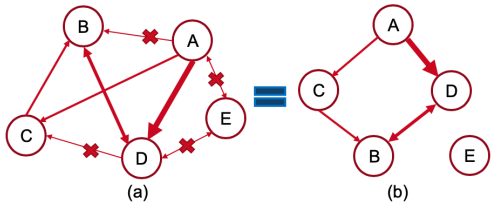

Figure 1: Schematic diagram of a direct relation graph (DRG). a) before dimension reduction; b) after dimension reduction.

The purpose of combustion modelling can vary and some species and reactions may be of no interest in particular cases. For example, a simulation which aims to predict peak combustor temperature is unlikely to need the solution of minor species such as NOx that are not on the main oxidation pathway. To take advantage of the additional dimension reduction that is possible, error propagation is applied to further trim the DRG. The dependency of a secondary species *P* from a primary species P_0 is quantified by

$$R_{P_0}(P) = \max_{\Omega}\left(\prod r_{ij}\right) \tag{8}$$

where Ω is all possible paths from P_0 to *P* and r_{ij} is defined by Eq. (11). Species with $R_{P_0}(P) < \varepsilon_{DAC}$ are also marked inactive and eliminated as illustrated schematically in Fig. 2. In practice multiple primary species, called the search initial set (SIS), are specified by the user depending on the aim of the modelling.

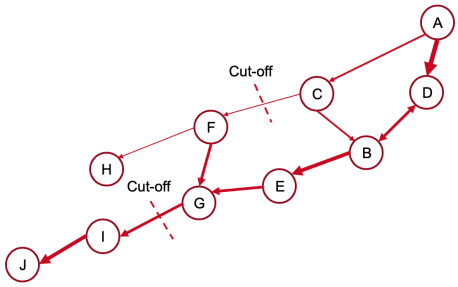

Figure 2: Schematic of a DRG with error propagation. $R_A(F) < \varepsilon_{DAC}$ and $R_A(I) < \varepsilon_{DAC}$

2.3 TDAC

TDAC combines ISAT and DAC. For each notional particle retrieval from the ISAT tree is attempted and if it fails DAC is employed to produce locally reduced kinetics, followed by DI to update the composition of the chemically active species. ISAT table entries are added and stored in complete composition space. However, the mapping gradient matrices are calculated by the reduced kinetics excluding the inactive species, and these result in more compact matrices. Therefore, the mapping gradient matrices are extended by unity elements corresponding to the inactive species. When the growing operation is called, the EOA grows in the active species direction of the original composition

point and the additional active species direction of the query point.

3 Validation case setup

TDAC has been implemented in an object orientated C++ open source environment and coupled with a stochastic turbulent chemistry solver called *mmcFoam* [4]. The TDAC code for stochastic particles has been adapted from the finite volume TDAC code in OpenFOAM v5. The performance of TDAC is tested by simulating a University of Sydney laboratory flame on a burner with adjustable inlet mixture homogeneity [15, 20]. Here we consider case FJ200-5GP-Lr300-59 which has a homogeneous mixture at the burner exit producing a non-premixed flame. This case was simulated previously by Galindo et al. using *mmcFoam* [7] and the same grid, notional particle distribution and boundary conditions have been used in the present work. Independent tests of ISAT with different values of ε_{ISAT} and DAC with different values of ε_{DAC} and SIS selections are conducted first. The ISAT tests use the 22-species DRM22 mechanism [21] and the DAC tests use the 53-species GRI-3.0 chemical mechanism [22]. Then computational efficiency of ISAT, DAC and the combined TDAC method are compared using DRM22, GRI-3.0 and the 111-species USC Mech II [23].

4 Results

All tests reported here involving ISAT have a constant value of N_d which determines when leaves are removed from the binary tree. In general, a smaller value requires less memory but leads to fewer retrievals, due to an overall smaller number of leaves and a less complete coverage by the tree of the accessed region in composition space, and increased need for DI and a corresponding increase in computational cost. A compromise is sought. Figure 3 shows the average distribution of the number of retrievals and the number of leaves against the number of timesteps a leaf has not been accessed, N_{na}, for a case with $\varepsilon_{ISAT} = 10^{-4}$ and $N_d = 200$. Over the range $0 < N_{na} < 20$ the number of leaves decreases by about one order of magnitude while the number of retrievals falls by about three orders of magnitude. Therefore, by using $N_d = 20$ we can achieve an 83% reduction in memory but only a 2% reduction in the number of retrievals relative to the case with $N_d = 200$. For consistency, all remaining tests use $N_d = 20$ although the optimum value would, in general, depend on ε_{ISAT} and the chemical mechanism being used.

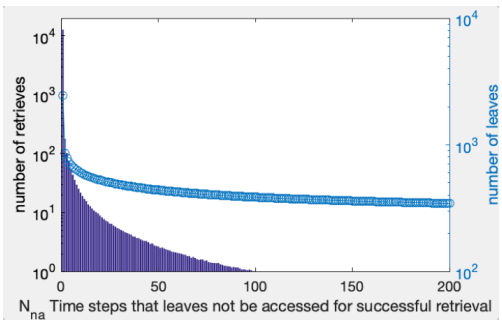

Figure 3: Distribution of the number of retrievals (left axis, purple bars) and the number of leaves (right axis blue circles) against the number timesteps that leaves have not been accessed. Result is for DRM22, $\varepsilon_{ISAT} = 10^{-4}$ and $N_d = 200$.

Figure 4 shows the effects of decreasing ε_{ISAT} tolerances, compared to DI, on the radial profiles of mean temperature and mass fractions of CO and OH. Results for $\varepsilon_{ISAT} = 10^{-3}$ have significant deviation from the DI curves, while there is rapid convergence for smaller values with $\varepsilon_{ISAT} = 10^{-4}$ and 10^{-5} producing results that are in excellent agreement with the DI results.

Figure 5 shows the convergence of DAC with decreasing ε_{DAC}. To examine sensitivity to the SIS selection, NO (which is not on the primary fuel oxidation pathway) is also shown. When CH_4 is the sole species in the SIS list, temperature CO and OH converge rapidly and $\varepsilon_{DAC} = 10^{-2}$ is seen to be sufficient. The convergence of NO is slower with $\varepsilon_{DAC} = 10^{-1}$ failing to produce any NO and $\varepsilon_{DAC} = 10^{-3}$ being required to produce results which are indistinguishable from the non-reduced (NR) results. However, when NO is added to the SIS list it is found that $\varepsilon_{DAC} = 10^{-2}$ is a sufficiently small value once again. The effect of ε_{DAC} and the SIS on the extent of dimension reduction is shown in Fig. 6 which plots the average proportion of notional particles in the ensemble versus the number of active species. For cases in which CH_4 is the sole species in the SIS list, the average number of active species increases significantly from around 26 to 36 when ε_{DAC} decreases from 10^{-2} to 10^{-3}. For the smaller tolerance it is observed that there are very few particles having less than 20 active species. When NO is added to the SIS list and $\varepsilon_{DAC} = 10^{-2}$ the average number of active species increases to about 32. A second smaller peak appears on the right hand side of the distribution corresponding to particles for which NO chemistry is evidently important.

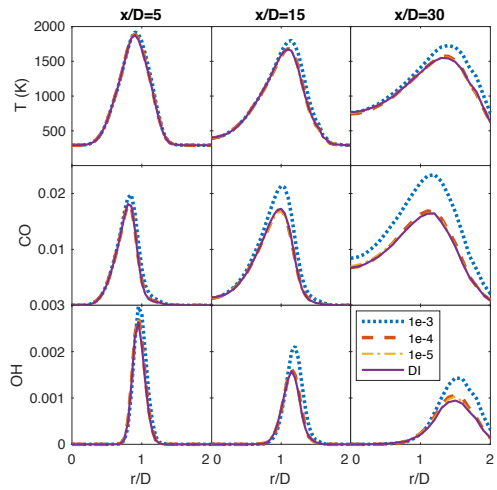

Figure 4: Radial profiles of mean temperature, CO and OH for different ε_{ISAT}. Result is for DRM22 and $N_d = 20$.

Finally, the speedup relative to DI of the non-reduced schemes is examined for ISAT, DAC and TDAC for three mechanisms with different numbers of species. The speedup factors, calculated by the ratio relative to the average time spent on chemistry using DI alone, are reported in Table 1. The results are for $\varepsilon_{ISAT} = 10^{-4}$, $\varepsilon_{DAC} = 10^{-2}$ and CH_4 as the sole species in the SIS list. The overall trend is similar to that reported in Ref. [12]. ISAT speedup increases at first when the number of species is increased from 22 to 53 but then the speedup deteriorates as the number of species increases to 111. DAC slows down simulation when small kinetics (DRM22) is used because of the overhead of reduction, however, as the number of species increases there is greater potential for DAC to dynamically eliminate reactions and species which make only minor contributions to the chemical solution and the speedup is enhanced for GRI-3.0 and USC Mech II as shown in Table 1. TDAC does not produce the highest speedup factors for any of these mechanisms. For DRM22, the efficiency drop relative to

ISAT is due to the cost of reduction. The situation is less clear for the two larger mechanisms and may be caused by the presence of frozen species which exacerbate the inhomogeneity of composition field, resulting in fewer retrievals. The speedup of all three methods is similar when USC mech II is used and future work will investigate performance for even larger chemical schemes.

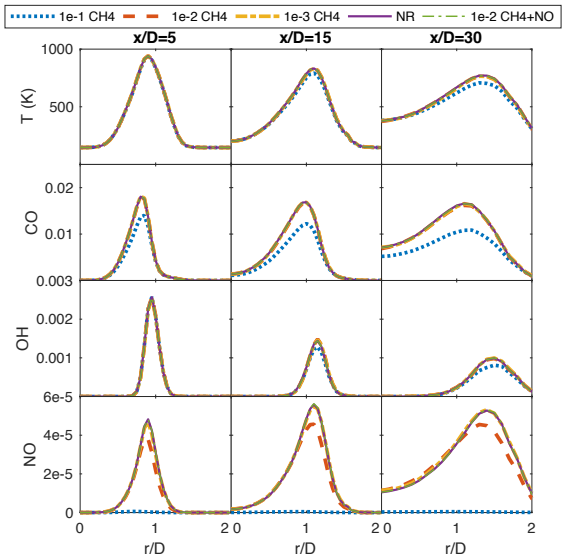

Figure 5: Radial profiles of mean temperature, CO, OH and NO for different ε_{DAC}. Result is for GRI-3.0.

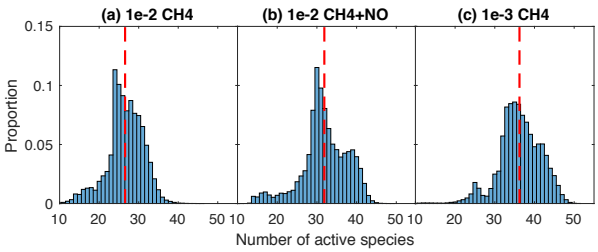

Figure 6: Average proportion of notional particles versus the number of active species for different ε_{DAC} and SIS. Red dashed line indicates the mean. Result is for GRI-3.0.

Mechanism	ISAT	DAC	TDAC
DRM22 (19)[a]	3.0	0.9	2.7
GRI-3.0 (53)[a]	3.5	1.6	1.6
USC-mech II (111)[a]	2.6	2.8	2.6

a. Number of species

Table 1: Chemistry integration speedup factor relative to DI. Results are $\varepsilon_{ISAT} = 10^{-4}$, $\varepsilon_{DAC} = 10^{-2}$ and CH_4 as the sole species in the SIS list.

5 Conclusions

The TDAC method for chemistry integration has been implemented into an open source stochastic turbulent combustion code called *mmcFoam* and tested by simulations of a homogeneous-inlet, non-premixed laboratory flame. Convergence of the ISAT and DAC parts of the method with decreasing values of ε_{ISAT} and ε_{DAC} has been demonstrated and sensitivity of DAC predictions with different primary species (SIS list) has been examined. Speedup factors relative to direct integration of all species have been reported for ISAT, DAC and TDAC for three different chemical mechanisms with varying number of species. DAC shows the highest speedup for the largest kinetics scheme while ISAT achieves the best result for the two smaller ones.

6 Acknowledgment

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